

SHORT BOUTS OF HIGH-INTENSITY RESISTANCE-STYLE TRAINING PRODUCE SIMILAR REDUCTIONS IN FASTING BLOOD GLUCOSE OF DIABETIC OFFSPRING AND CONTROLS

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ABSTRACT

Russell, RD, Nelson, AG, and Kraemer, RR. Short bouts of high-intensity resistance-style training produce similar reductions in fasting blood glucose of diabetic offspring and controls. *J Strength Cond Res* 28(10): 2760–2767, 2014—Family history of diabetes (FH) is associated with impaired cardiometabolic function. Aerobic exercise improves insulin sensitivity, though resistance training studies on fasting glucose (FG) in FH are lacking. This study examined the effects of 7 weeks of high-intensity-resistance-focused training (HIRFT), including circuit, core, and plyometric resistance training on FG in FH and matched controls (CON). We hypothesized that HIRFT would reduce FG levels, with greater reductions in CON. Thirty-eight healthy men and women (23.5 ± 2 years; 171 ± 7.4 cm; 71 ± 14 kg) participated in 7 weeks of HIRFT including full-body, plyometric, and core resistance training on alternate days. Fasting glucose was analyzed before and after the 7-week training before and after workouts. One repetition maximum was calculated for bench press, squat, and deadlift before and after training. Body mass index and resting HR remained unchanged. Fasting glucose declined similarly between groups with training (-0.23 ± 0.08 vs. -0.20 ± 0.07 mmol·L⁻¹, $p < 0.01$ for FH and CON, respectively), whereas strength increased (kg) (bench: 8.0 ± 1.8 , squat: 19.4 ± 4.6 , deadlift: 16.4 ± 3.6 , overall mean percent increase: 38.9 ± 9.2 , $p < 0.001$). Ten-minute post-exercise glucose decreased (-0.65 mmol·L⁻¹, $p = 0.05$) with training, with no differences between groups. Changes in FG and strength increase were inversely correlated ($r = -0.519$, $p = 0.05$). Strength increased equally between groups. Data indi-

cate that HIRFT reduces FG concentrations similarly in FH and CON, making it effective for improving FG in FH.

KEY WORDS resistance training, fasting glucose, diabetes family history

INTRODUCTION

Insulin resistance (IR) precedes type 2 diabetes (T2D) and is the best predictor for future development of T2D (25). Because IR is related to genetic variation (53) and comorbidities, including elevated lipid levels (6), metabolic inflexibility (8), and impaired glucoregulation (52), studies traditionally focus on these comorbidities, making it difficult to distinguish the initial trigger for IR. Offspring of people with T2D (family history of T2D and FH) have a greater lifetime risk of developing T2D than those with no family history (CON) (37,51) and have impaired hexokinase II activity and peroxisome proliferator-activated receptor gamma coactivator 1 (PGC1) expression similar to T2D (32,41), which leads to decreased expression of nuclear respiratory factor 1 (NRF-1) (32) and impaired mitochondrial respiration, all of which may help explain the impaired metabolic function in this population (38). The benefit of studying metabolic function in healthy FH is that these individuals have not yet developed confounding factors associated with T2D, like glucotoxicity or obesity, yet they display impaired metabolic flexibility similar to T2D (38). This suggests that the pathogenesis of T2D can be examined at its earliest time points in FH (29). Research has been limited to sedentary, IR FH only, making it difficult to isolate the pathologic connection between FH and T2D leading to IR and diabetes (3,50,51). Although some investigations have revealed that aerobic exercise training in FH increases metabolic flexibility, glucoregulatory function, and insulin sensitivity index to levels similar to sedentary CON (1,3), this may be due to FH already having impaired insulin sensitivity at baseline (3). Thus, it is important to not only identify specific metabolic perturbations in this group but

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also to determine what types of interventions would be most effective in the prevention and treatment of cardiometabolic disease.

Exercise intervention enhances insulin sensitivity and is associated with reduced whole-body fasting respiratory quotient in obese insulin-resistant individuals (15) and improved metabolic flexibility in T2D (27) and insulin-resistant FH (3). Exercise, which decreases with increased age (13), is the only treatment that improves whole body and skeletal muscle oxidative capacity (33); however, there is disagreement over the form of exercise most effective in T2D prevention and treatment (17), and such interventions in insulin sensitive FH are lacking.

Current evidence suggests that resistance exercise aids in the prevention and treatment of metabolic disease because it (a) increases insulin sensitivity (lowers HbA1c and hyperglycemia by up to 24%) in certain diseased states (2,20), (b) improves insulin action by enlarging skeletal muscle mass (46), and (c) recruits the fiber type with the lowest insulin sensitivity, type IIX (44). Furthermore, resistance training increases lean body mass, resting energy expenditure (28), fat oxidation (31), and muscle oxidative capacity (47), which may prevent increased fat mass (26). Aerobic and resistance exercises have similar lowering effects on fasting blood glucose and improvements in insulin sensitivity, yet combination of the 2 has additive effects (22). However, in addition to improving IR, very long bouts of aerobic exercise acutely increase ghrelin and peptide yy-36 (PYY) (39), whereas intense resistance training has been shown to have the opposite effect (16). Furthermore, alternative forms of resistance-style training such as kettlebell exercise have recently been shown to elicit the same metabolic responses as moderate-intensity graded treadmill walking (48). Thus, in the last 8 years, high-intensity aerobic training has gained recognition as a means to improve IR, glycemic control, and muscle oxidative capacity (7,10,16,19,24). Therefore, this study determined the clinical benefits of high-intensity-resistance-focused exercise training (HIRFT) in healthy young people with (FH) and without (CON) a family history of T2D, with identical baseline fasting glucose (FG) levels. This training program used fast-paced high-intensity resistance-type exercise 5 days per week with alternating muscle focus on different days, for example, full-body weight training on Monday and Friday, core body workouts using resistance on Tuesday and Thursday, and explosive plyometric movements on Wednesdays (Table 1). It is likely that this kind of fast workout will have better long-term adherence because the time commitment is low.

Because we have observed metabolic inflexibility in young healthy FH (38), and others observed that metabolic inflexibility is directly related to fasting blood glucose (14), we chose fasting blood glucose as a marker of metabolic impact from this training program. We hypothesized that HIRFT would reduce fasting blood glucose, but contrary to a previous study where groups were not matched at baseline (3), the reductions would be more

pronounced in the CON group because both groups were matched for FG.

METHODS

Experimental Approach to the Problem

The study consisted of 3 main parts: (a) a pretraining test day including fasting blood glucose measurement followed by determination of an estimated 1 repetition maximum (RM) for bench press, squats, and deadlift, and 2 postworkout blood glucose measurements; (b) 7 weeks of short HIRFT using fast-paced superset circuit training, body core, and plyometrics training; and (c) a posttraining test day including fasting and postexercise blood glucose and reevaluation of calculated maximal strength. Participants were instructed neither to perform strenuous exercise nor consume alcohol or caffeine at least 2 days before either test day. Fasting blood glucose was measured to establish at baseline to ensure that the participants are evenly matched and not prediabetic. To determine if HIRFT is an effective means of resistance training, a subgroup was randomized to perform traditional multiset resistance training instead of the fast-paced circuits, and strength gains were compared. Fasting glucose was remeasured after an acute bout of HIRFT and after 10 minutes of passive recovery to determine glucose appearance and clearance rates. Furthermore, fasting, postexercise, and 10-minute postexercise glucose was reassessed after 7 weeks of HIRFT to determine if this style of training affects FG, exercise-induced glucose spikes, and glucose clearance rates. Postexercise lactate was measured to better quantify exercise intensity before and after HIRFT intervention. Though specific mechanisms of cardiometabolic function are not being measured, the data generated from this study were able to validate HIRFT, and help direct future research in this at-risk population, and lifestyle modifications to prevent IR and T2D.

Subjects

Thirty-eight subjects (18 FH and 20 CON, 18–34 years) were recruited from the Baton Rouge, LA, area for participation in a 7-week study. The investigation determined the effects of a HIRFT program consisting of circuit training, body core training, and plyometrics training on different days, 5 days per week, on strength gains and FG. Subjects enrolled through an interactive online system that described study, but did not mention the 2 methods. Ten of the subjects were then randomized by the automated computer system into a different training module (5 CON and 5 FH) that substituted traditional resistance training for circuit training to validate the 10-minute HIRFT resistance workouts with traditional resistance training. Family history of diabetes was categorized as being a first or second degree relative of a person with T2D. Age, gender, body mass index (BMI), and physical activity were matched between FH and CON. The purpose, potential risks, and benefits of participation in this study were fully explained to each participant,

TABLE 1. Weekly training schedule.

Days	Monday	Tuesday	Wednesday	Thursday	Friday
Training mode	Resistance/circuit training	Resistance-type core	Plyometrics	Resistance-type core	Resistance/circuit training
Duration	40/10 min	10–15 min	40 min of 10-s bouts	10–15 min	40/10 min

and a signed, written consent was obtained before enrollment. The study was approved by the Ethical Committee of Louisiana State University (LSU office of IRB). Inclusion criteria for FH and CON groups were no overt disease, sedentary, fasting blood glucose $<6.67 \text{ mmol}\cdot\text{L}^{-1}$, and exclusion criterion were fasting blood glucose $>6.67 \text{ mmol}\cdot\text{L}^{-1}$, participating in more than 1 hour of physical activity in a given week, BMI $>25 \text{ kg}\cdot\text{m}^{-2}$, taking prescription medication other than oral contraceptives, or a previous diagnosis of T2D or prediabetes. A physical activity questionnaire was completed before participation in the study to determine physical activity levels. Subject characteristics are summarized in Table 2.

Procedures

Blood Glucose. After a 10-hour fast, blood glucose was measured using a validated glucometer (Accu-Chek Compact Plus, Indianapolis, IN, USA) (49) in duplicate to better insure accuracy. Blood glucose measurements were taken in duplicate after 10 minutes of rest, and then each participant was given a 110 kcal breakfast food bar before exercise. Blood glucose was reevaluated in duplicate immediately after (time 0) and 10 minutes after (time 10) exercise for both test days. Participants recorded their diet the day before the first test day and repeated the same diet on the day preceding the posttraining test day. In addition to strength progression and muscular failure during lifts, blood lactate was also measured at the same time as blood glucose using a lactate monitor (Lactate Pro, Quesnel, BC, Canada) to better quantify exercise intensity/metabolic stress. Participants were asked to record their diet from the day before the first test day and repeat it the day before post-training testing.

Exercise Training. Training was performed 5 days per week at the same time every day in the LSU Strength and Conditioning facility. This is a 10,000 square foot facility with 28 multipurpose power stations, 36 assorted selectorized machines, and 10 dumbbell stations along with a plyometric specific area (with medicine balls, hurdles, plyometric boxes, and assorted speed and agility equipment). Two treadmills, 4 stationary bikes, 2 elliptical cross trainers, and 1 stepper and stepmill were used for warm-ups and cool-down. The weekly training used mostly free weights and

included weight lifting on Monday and Friday, core body work on Tuesday and Thursday, and plyometrics on Wednesday (Table 1). The first of each workout (weight-training, core, and plyometric) was used to illustrate how each exercise was performed and gave each participant the opportunity to optimize their starting load for each exercise. Ten participants were randomized into a subgroup of traditional 2-set resistance training with the second set going to complete muscular fatigue to validate strength gains from HIRFT with traditional multiset resistance training. Individual resistance exercises were the same between both modes of training and included squat, bench press, lateral pull-down/seated row, shoulder press, push-up, bicep curl, triceps extension, and deadlift exercises performed in that order. Time to complete each session of multiset training was 40 minutes, including 1 minute rest between sets, whereas each circuit-training session was limited to 10 minutes excluding warm-up and cool-down with no rest between exercises. All subjects completed the same plyometric and body core exercises. Core and plyometrics continually changed with increased fitness. Core was performed in <15 minutes, using weighted/resistance techniques including, but not limited to dumbbell sit-ups, medicine ball toss, rolling on ab wheel (or barbell), and plank positions. Plyometric workouts were completed in 50 minutes and used short bouts of explosive movements with several minutes of rest between. Some workouts included stadium sprints, clapping push-ups, box jumps, obstacle hops, ramp runs, and various fast medicine ball and dumbbell movements. Participants were encouraged to drink water ad libitum before, during, and after workouts. Training progression was continuously monitored and load adjusted for all participants in both exercise groups to ensure continued progression, enabling all participants to continue to work out at 65–85% of 1RM throughout training. One participant from the HIRFT group was excluded for noncompliance. Both groups had similar ratios for gender and family history, and began training on the same day an hour apart, and performed the same exercises. The circuit-training group was reached muscular fatigue within 8–12 repetitions at 65–85% of calculated 1RM on their 1 set, as opposed to the multiset group, which reached muscular fatigue on the second set.

TABLE 2. Subject characteristics (mean $\pm$ SD).*

	FH	CON
Age (yrs)	23.0 $\pm$ 4.4	24.1 $\pm$ 8.4
Height (cm)	170.5 $\pm$ 2.9	173.8 $\pm$ 1.9
Weight (kg)	70.6 $\pm$ 12.8	73.0 $\pm$ 13.5
BMI (kg $\cdot$ m ⁻²)	24.3 $\pm$ 0.24	24.2 $\pm$ 0.21

*BMI = body mass index.

Analyses. Strength assessment was conducted using a modified Epley formula (11):

$$1RM = \left(\left[\frac{r}{30} \right] + 1 \right) \cdot w,$$

where r = repetitions and w = weight lifted. This formula is highly correlated with actual 1RM at $r = 0.99$ (35). After a thorough warm-up, each person attempted 1 set of bench press, squats, and deadlift to complete muscular failure and then used the formula above to estimate 1RM. To ensure accurate 1RM estimation, participants were required to perform testing lifts with weight that resulted in fatigue within 8 repetitions. Blood glucose and lactate concentrations were assessed by finger prick as described above.

Statistical Analyses

All statistics were calculated using SPSS 17 (SPSS, Inc., Somers, NY, USA) and Stata SE software version 11 (StataCorp, College Station, TX, USA). One-way analysis of variance (ANOVA) was used to test for differences between

TABLE 3. Pre- to posttraining-induced changes in strength (kg $\pm$ SD).*†

	FH	CON	p
% strength	56.8 $\pm$ 187	42.4 $\pm$ 139	0.0001
Bench press (kg)	15.6 $\pm$ 10.8	8.8 $\pm$ 8.9	0.0001
Squat (kg)	32.2 $\pm$ 24	23.7 $\pm$ 18	0.0001
Deadlift (kg)	20.7 $\pm$ 15.6	17.4 $\pm$ 15.6	0.0001

* p = overall group changes from pre- to posttraining.

†No differences were noted between groups.

subjects at baseline. Repeated-measure ANOVA was used to test for training $\times$ time effects on strength and blood glucose and lactate. In all appropriate analysis, family history, BMI, sex, and training mode were included as a between-subject factor. Pearson product-moment correlations were used to determine relationships between variables, including family history category, training progression (changes in strength), changes in blood glucose, and changes in blood lactate. All comparisons were accepted as significant at the alpha level $p \leq 0.05$, and p values are defined for all variables.

Power calculations were performed based on previous resistance training trials in a similar population. We expected to have complete data on a minimum of 20 subjects (10 per group per group) after allowing for data loss and attrition. Power calculations were conservatively based on these numbers. All calculations assumed a 2-sided significance level of $\alpha = 0.05$ and, where applicable, a moderate

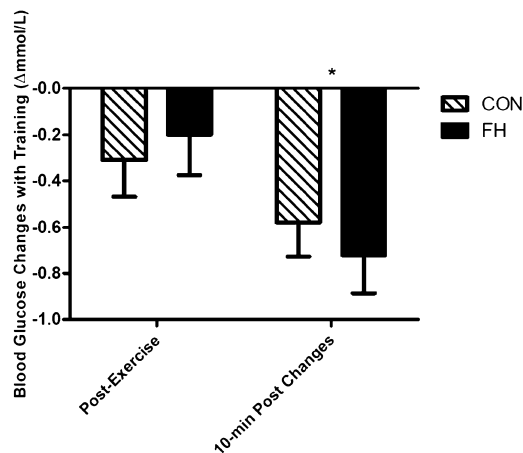


Figure 1. Pre- to posttraining blood glucose changes (Δ mmol $\cdot$ L⁻¹) between FH and CON immediately after and 10 minutes after acute exercise (post - pre). * $p < 0.05$.

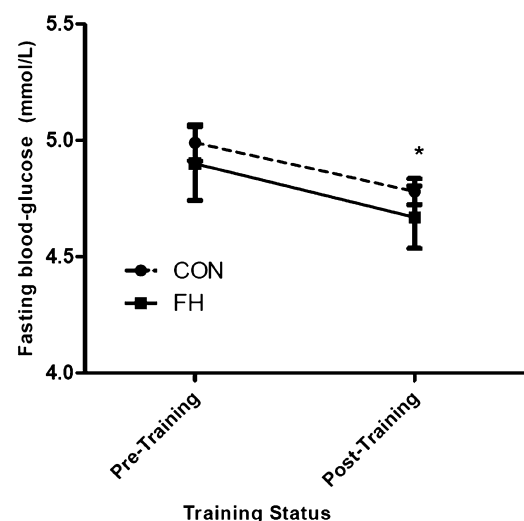


Figure 2. Fasting blood glucose concentrations before and after 7-week resistance training (mmol $\cdot$ L⁻¹). * $p < 0.05$.

correlation of $r = 0.5$ between repeated measurements within subjects. Contrasts were evaluated after significant omnibus tests to minimize type I error rates. Using the whole sample, our design had 80% power at significance level $\alpha = 0.05$ to detect an overall interaction between time and FH group with a small effect size of Cohen's $f = 0.20$, which corresponds to a change from no difference at baseline to a difference of $d = 0.50$ between the most extreme changes with intervention. The detectable effect sizes presented here are in the small to medium range using the conventions of Cohen's 87 who defined $f = 0.25$ and $d = 0.5$ as a standard medium effects. In addition, our minimum detectable effect sizes compare favorably to results from other studies and our own preliminary data.

RESULTS

There were no baseline differences in fasting blood glucose or calculated maximal strength between FH and CON. Both training regimes resulted in similar strength gains and blood glucose changes. Though the specific HIRT circuit training showed significant reductions in fasting blood glucose ($p \leq 0.05$), both training groups were combined to maximize power for all parameters. Fasting, postexercise, and 10-minute recovery blood glucose concentrations were not different between FH and CON (Figure 1). Strength increased from pre- to posttraining similarly in both groups (Table 3), whereas fasting blood glucose concentrations decreased with training overall ($p = 0.0054$), with no differences between FH and CON groups ($p = 0.7$) (Figure 2). Blood glucose concentrations immediately after acute exercise did not change with training. However, after 10-minute passive recovery, glucose concentrations declined similarly ($p = 0.05$) in FH and CON groups with training (-0.72 ± 0.7 and $-0.58 \pm 0.74 \text{ mmol} \cdot \text{L}^{-1}$, respectively) (Figure 1). There was also an inverse correlation between percent strength gains and decreased blood glucose concentrations

from pre- to posttraining ($r = -0.519$, $p = 0.05$). Moreover, acute exercise recovery lactate (after 10 minutes of passive recovery) was lower after training ($p \leq 0.05$) in both groups (Figure 3).

DISCUSSION

There were no differences between FH and CON in pretraining strength, BMI, or blood glucose or lactate before or after exercise. The hypothesis that both modes of training would illicit the same strength gains was supported; moreover, they also yielded the same FG and lactate responses over time. Therefore, although the fasting blood glucose reductions with HIRT were already significant, data were combined from both modes of training to increase statistical power for glucose and lactate changes. The hypothesis that FG would decrease more in CON than FH was not supported, though fasting blood glucose decreased in CON and FH alike with training ($p < 0.01$). This is the first study demonstrating that blood glucose concentrations are reduced similarly in FH and CON with HIRFT. In addition, there were no differences in glucose or lactate variables between men and women. This is in agreement with our previous work that detected impaired metabolic flexibility in FH (38), also with no gender differences.

Both modes of training were combined for statistical analysis for glucose and lactate. A high-volume load for strength training was used to increase potential exercise benefits, specifically because previous research indicates that 8 weeks of low-intensity low-volume resistance is not sufficient to decrease metabolic risk factors in T2D (18). In contrast, Bacchi et al. (2) show short-term (3 times per week for 10 weeks) resistance training significantly reduces HbA1c and improves 48-hour glycemic variation as measured by continuous glucose monitoring system. Furthermore, high-intensity interval training has been shown to improve homeostatic model of insulin resistance in individuals with T2D in as little as 2 weeks (40). Our HIRFT protocol involves HIRFT 5 days per week, alternating workouts. Thus, the total number of workouts performed in this study was higher, making the relatively short training interval more effective. Finally, rodent models indicate that a combination of aerobic and resistance training can increase phosphorylated p70S6K, a marker of mTORC1, after only 1 bout of combined training, indicating the activation of pathways leading to muscle synthesis (30). Resistance training is associated with increased strength and increased muscle mass as well as improved whole-body glucose disposal rate (12), which would help explain the reductions of FG and post-recovery lactate in this study. However, we were unable to determine whether glucose disposal occurred because of increased microvascular nutritive flow or increased muscle mass as a result of HIRFT. In this study, strength increased on average 51.2% with training overall. Furthermore, a recent study indicated that similar resistance-type circuit training can decrease metabolic risk factors in young overweight males (9). The increased strength observed in this study suggests that the

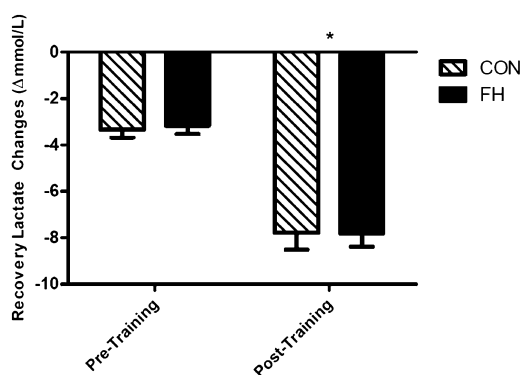


Figure 3. Changes in 10-minute recovery blood lactate concentrations before (pre) and after (post) resistance training between FH groups. * $p < 0.05$.

stimulus used for resistance training was sufficient to elicit a physiologic adaptation of muscle; however, lean muscle mass was not specifically evaluated in this study.

Fasting blood glucose decreased with training. There were no differences in the FG reductions between groups with training. This similar glucose reduction between groups indicates that although the FH population (recruited from the same pool as this study) has impaired metabolic flexibility independent of any glycemic problems (38), their glucoregulatory function seems to adapt to resistance exercise training in a similar fashion as CON. The overall decline in fasting blood glucose concentrations indicates tighter glucoregulatory control after training. This may be because of greater glucose uptake in muscle as previously indicated (12). However, reduced hepatic glucose release cannot be ruled out because we did not measure the glucose source specifically. Therefore, it was not possible to distinguish between glucose appearance and removal rates, and thus we only described what is acutely found in the blood at specific time points. However, postexercise blood glucose decreased more dramatically during passive recovery after training in both groups, indicating greater glucose removal with training. Nonetheless, whether a fasting decrease was from reduced appearance or increased removal, a tighter blood glucose regulation exists after training, further supporting previous studies that this type of training reduces IR (4,9) independent of body mass changes. Recent studies indicate that microvascular IR precedes conduit artery and muscle IR and contributes to the progression of cardiometabolic syndrome (34). If decreased glucose concentration during passive recovery seen after training is due to increased glucose removal, it is possible that training increases nutritive routes of microvascular recruitment, leading to greater myocyte perfusion and glucose uptake. This would not only explain why FG decreased with training but would also help explain the faster postexercise recovery time as indicated by more dramatic 10-minute reductions in both glucose and lactate. More work needs to be performed to examine the effects of resistance training on microvascular recruitment, especially in the FH population.

In this study, there were no changes in fasting lactate; however, sharp increases in lactate in both groups were observed immediately after exercise before and after 7 weeks of training. However, this acute postexercise lactate increase was higher after training than pretraining (9.2 ± 0.7 vs. 11.9 ± 0.8 , $p = 0.034$), indicative of training adaptations (36). In conjunction with increased estimated 1RM, this increased lactate indicates that a greater absolute workload was achieved after 7 weeks of training without increasing the relative workload. This may be because of greater muscle mass with training, though no increases in body weight were noted. Though blood lactate was higher immediately after acute exercise with training, there were no differences in blood lactate after 10 minutes of passive recovery from pre- to posttraining. This indicates a greater rate of lactate removal with training (Figure 3). This

greater decrease in blood lactate concentrations may be an indication of higher oxidative enzyme capacity with training, as much lactate is removed by oxidative muscle fibers adjacent to lactate-producing glycolytic fibers and by the liver (5,42).

Simoneau and Kelley (43) show that an increased ratio of glycolytic to oxidative enzymes contributes to IR in skeletal muscle of patients with T2D. Granted, this study was not designed to monitor lactate removal or to determine whether the greater decrease with passive recovery was due to less lactate appearance, greater rate of removal, or a combination of both. It is plausible that lactate removal occurred at a faster rate after training. Because exercise upregulates mitochondrial biogenesis transcription factors such as PGC-1 α (23) and the various ways in which lactate can be removed, including by oxidative muscle fibers (5), it is plausible to expect that HIRFT in this study elicited an upregulation of PGC-1 α leading to increased oxidative muscle fibers and higher mitochondrial content, and tighter glucoregulatory control. This is supported by previous studies indicating that high-intensity interval training (21), and combined aerobic and resistance training (45), improves skeletal muscle mitochondrial content and muscle metabolism. Nevertheless, the improved recovery as seen by reduced lactate 10 minutes after exercise further supports the notion of improved microvascular nutritive flow.

PRACTICAL APPLICATIONS

Although no prescreening testing for metabolic flexibility was performed, FH is at much higher risk of developing T2D (51) and has impaired metabolic flexibility equal to those with T2D during an oral glucose challenge (38). Therefore, we fully expect the FH group in this study to have the same metabolic impairments. This is the first study to demonstrate that HIRFT, including resistance circuits, core, and plyometrics, as well as traditional multiset resistance training with core and plyometrics are equally effective in reducing blood glucose and increasing strength in relatively inactive young adults with and without a family history of T2D equally. Moreover, the inverse correlation between strength gains and reduction in blood glucose in FH suggests that HIRFT could be an effective form of exercise medicine to prevent the development of T2D in this population. Of particular importance in studying the FH population is that they lack many of the confounding factors that are linked with diabetes and prediabetes, yet are predisposed to metabolic impairment (38). Not only does this research directly apply to the FH population but it also helps isolate the earliest detectable challenges in metabolic function that may be responsible for the development of T2D. In addition, it is likely that the short-time commitment for this kind of exercise may lead to better adherence of training over time, resulting in better overall cardiometabolic health. Further research is needed to determine specific mechanisms whereby resistance training alters cardiometabolic health in the FH population.

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